Introduction to Camera Calibration

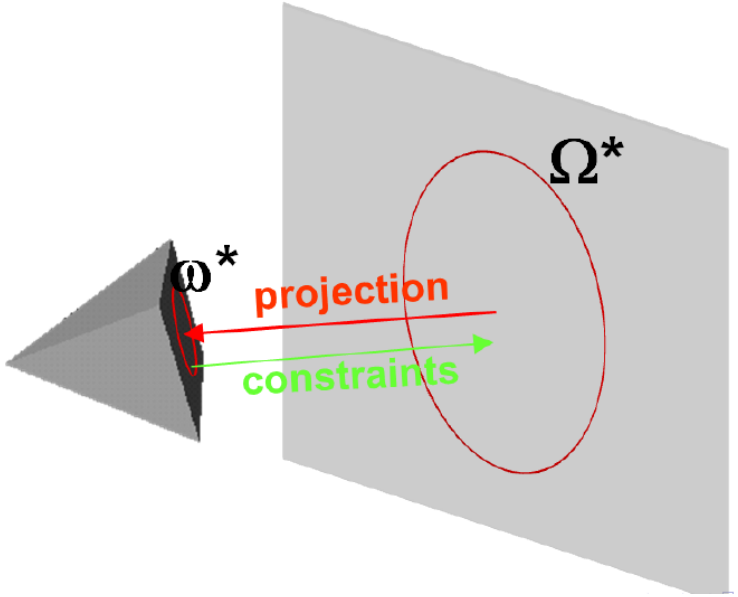
New Slides 27/9/2026

Chapter 11

Camera Calibration

- How to estimate the intrinsic and extrinsic parameters of a camera?

- $\vec{x}_{image\ plane\ 3\times1} = P_{3\times4}\ \vec{X}_{World,4\times1}$
  - Assumptions:
    - Pairs of $\vec{x}_{image\ plane\ 3\times1}$ and $\vec{X}_{World,4\times1}$ are Known
    - Camera Matrix $P_{3\times4} = K\left[R|\vec{t}\right] = KR\left[I\middle|-\tilde{C}\right]$ is Unknown
      - Two Parts: Intrinsic K and Extrinsic $R\ and\ \vec{t}$



- First focus on K
  - Let's utilize the information about:
    - Absolute Conic $\Omega_\infty = I_{3\times3} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$:
      - Intersection of a sphere with $\pi_\infty = [0 \quad 0 \quad 0 \quad 1]^T$. This is a circle or a conic.
    - Circular Points: $I = \begin{bmatrix}1\\ i\\ 0\end{bmatrix}\ and\ J = \begin{bmatrix}1\\ -i\\ 0\end{bmatrix}$, where $i = \sqrt{-1}$
      - Intersection of a circle (i.e., Absolute Conic) in a plane with with $l_\infty = [0 \quad 0 \quad 1]^T$
      - So I and J are located on Ω_∞

Recall

– The equation of the point conic C

- $\vec{x}_{1\times3}^{T} C_{3\times3} \vec{x}_{3\times1} = 0$

– How a conic C transforms via a Homography H to a conic C'

$$C' = H^{-T} C H^{-1}$$

– $C = \begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}$, 5 DoF $(Assume, f = 1)$

- So, the circular points I and J will have:
  - $\vec{I}^T_{1\times3}\Omega_\infty\vec{I}_{3\times1} = 0$, and $\vec{J}^T_{1\times3}\Omega_\infty\vec{J}_{3\times1} = 0$
  - And, any 3×1 point $\vec{x}_d$ (H.C.) located on Ω_∞ will have:
    - $\vec{x}_d^T\Omega_\infty\vec{x}_d = 0$

- Ω_∞ can be transformed via $\boldsymbol{H}$ to another conic (will be called **image of absolute conic $\boldsymbol{w}$**.
  - $\mathrm{C}' = H^{-T}CH^{-1}$, $w = H^{-T}\Omega_\infty H^{-1}$, $\Omega_\infty = I_{3\times3} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$
  - $w = H^{-T}\ H^{-1}$

$$\vec{x}^{\surd}_{image\ plane} = P^{???}_{3\times4}\ \vec{X}^{\surd}_{World}$$

- Since, we talk about the absolute conic, let's consider $\vec{X}^{\surd}_{World,\infty}$ (ideal point)
  - $\vec{X}^{\surd}_{World,\infty} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ 0 \end{bmatrix} = \begin{bmatrix} \vec{x}_d \\ 0 \end{bmatrix}$, $\vec{x}_d$ is 3×1 directional vector of the point
    - **Consider it a fact: $\vec{x}_d$ is located on Ω_∞**
  - Let's now try capture an image (forward projection) of $\vec{X}^{\surd}_{World,\infty}$ using $P^{???}_{3\times4}$
    - $\vec{x}^{\surd}_{image\ plane} = P^{???}_{3\times4}\ \vec{X}^{\surd}_{World,\infty} = KR\left[I\middle|-\tilde{C}\right]\begin{bmatrix} \vec{x}_d \\ 0 \end{bmatrix} = KR\ \vec{x}_d$

$$\vec{x}^{\surd}_{image\ plane} = KR\ \vec{x}_d$$

- Note:
  - K is non-singular and upper triangle
  - R is orthonormal
  - Thus: KR is non-singular 3×3 matrix which is a homography
    - Let $H = KR$
  - $\vec{x}^{\surd}_{image\ plane} = H\ \vec{x}_d$

Recall

- Ω_∞ can be transformed via $H = KR$ to another conic (will be called image of absolute conic w.

  - $C' = H^{-T} C H^{-1}, w = H^{-T} \Omega_\infty H^{-1}, \Omega_\infty = I_{3\times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

  - $w = H^{-T}\ H^{-1}$

  - $w = (KR)^{-T}\ (KR)^{-1} = K^{-T}\ R^{-T} R^{-1}\ K^{-1} = K^{-T}\ K^{-1}$

    - Thus, if we know w, then we would know K

- **Note that $\omega = K^{-T}K^{-1}$ is the very real image of some something very unreal.**

- Recall that $K = \begin{bmatrix} \alpha_x & 0 & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow K^{-1} = \begin{bmatrix} 1/\alpha_x & 0 & -x_0/\alpha_x \\ 0 & 1/\alpha_y & -y_0/\alpha_y \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow K^{-T} = \begin{bmatrix} 1/\alpha_x & 0 & 0 \\ 0 & 1/\alpha_y & 0 \\ -x_0/\alpha_x & -y_0/\alpha_y & 1 \end{bmatrix}$

- $\omega = K^{-T}K^{-1} = \begin{bmatrix} 1/\alpha_x & 0 & 0 \\ 0 & 1/\alpha_y & 0 \\ -x_0/\alpha_x & -y_0/\alpha_y & 1 \end{bmatrix} \begin{bmatrix} 1/\alpha_x & 0 & -x_0/\alpha_x \\ 0 & 1/\alpha_y & -y_0/\alpha_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\alpha_x^2} & 0 & \frac{-x_0}{\alpha_x^2} \\ 0 & \frac{1}{\alpha_y^2} & \frac{-y_0}{\alpha_y^2} \\ \frac{-x_0}{\alpha_x^2} & \frac{-y_0}{\alpha_y^2} & \frac{x_0^2}{\alpha_x^2} + \frac{y_0^2}{\alpha_y^2} + 1 \end{bmatrix}$

How to know w

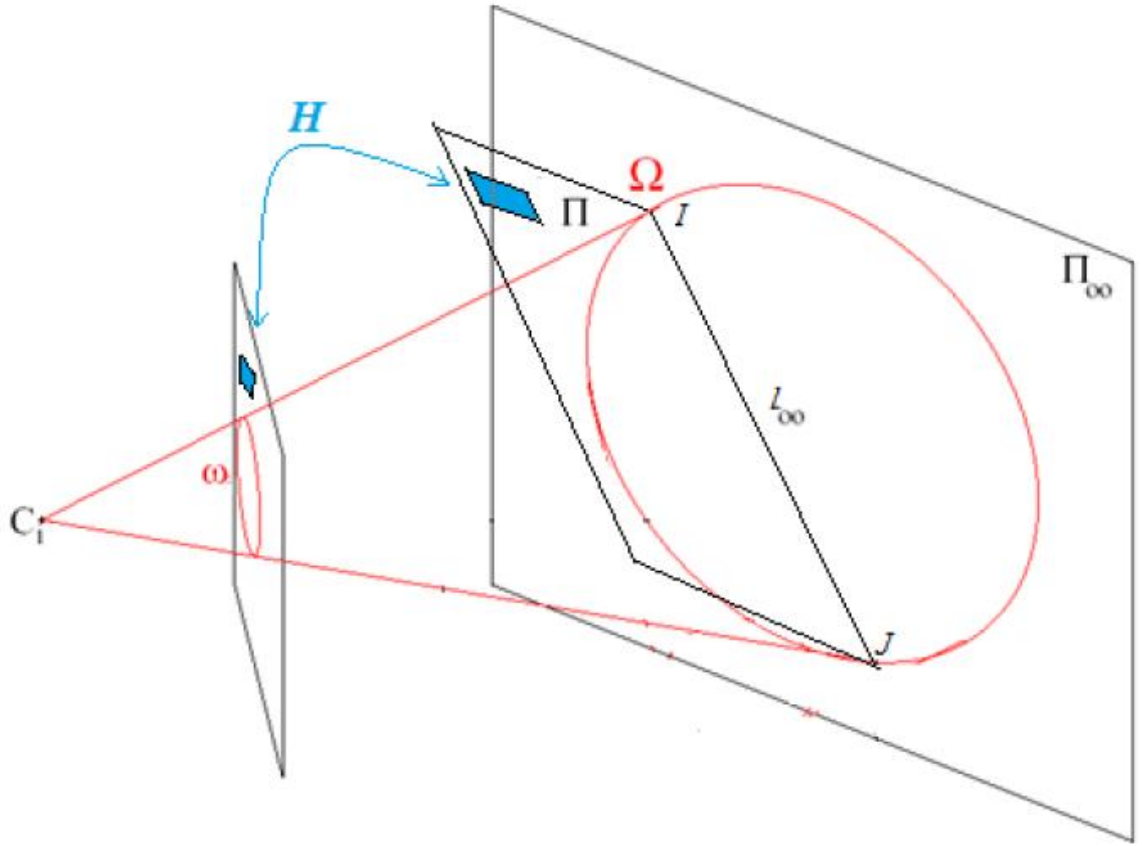
- w is a conic in the image plane of the camera and has 5 DoF
  - Thus, need five equations to solve for these unknown 5 DoF
  - How to find such equations?
    - Let's utilize the points located on w
      - We have two points already
        - » $\vec{x}_1$: Image point of the first circular point $I = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$, where $\vec{x}_1^T w \vec{x}_1 = 0$
        - » $\vec{x}_2$: Image point of the second circular point $J = \begin{bmatrix} 1 \\ -i \\ 0 \end{bmatrix}$, where $\vec{x}_2^T w \vec{x}_2 = 0$

$$\vec{x}^{\surd}_{image\ plane} = H^{???}_{3\times3}\ \vec{x}^{\surd}_{d\ 3\times1}$$

- $\vec{x}^{\surd}_{d\ 3\times1}$ is known
  - because we know $I = \begin{bmatrix}1\\ i\\ 0\end{bmatrix}$ and $J = \begin{bmatrix}1\\ -i\\ 0\end{bmatrix}$ are indeed located on Ω_∞
- $\vec{x}^{\surd}_{image\ plane}$ is also known since we can estimate the unknown H? How?
  - See previous slide.

Use of a Calibration Pattern

- H can be easily estimated using a probing plane



- Let the estimated $H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} = \begin{bmatrix} \vec{h}_1 & \vec{h}_2 & \vec{h}_3 \end{bmatrix}$

- Image of the first circular point (imaginary):

$$\vec{x}_1 = HI = \begin{bmatrix} \vec{h}_1 & \vec{h}_2 & \vec{h}_3 \end{bmatrix} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = \vec{h}_1 + i\vec{h}_2$$

- Image of the second circular point (imaginary):

$$\vec{x}_2 = HJ = \begin{bmatrix} \vec{h}_1 & \vec{h}_2 & \vec{h}_3 \end{bmatrix} \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix} = \vec{h}_1 - i\vec{h}_2$$

- Let’s now use:
  - $\vec{x}_1$: Image point of the first circular point $I = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$, where

    $\vec{x}_1^T w \vec{x}_1 = 0$

  - $\vec{x}_2$: Image point of the second circular point $J = \begin{bmatrix} 1 \\ -i \\ 0 \end{bmatrix}$, where

    $\vec{x}_2^T w \vec{x}_2 = 0$

The image of the Absolute Conic Ω_∞: ω

- $\vec{x}_1^T \omega \vec{x}_1 = 0 \Rightarrow \left(\vec{h}_1 + i\vec{h}_2\right)^T \omega \left(\vec{h}_1 + i\vec{h}_2\right) = 0$, and
- $\vec{x}_2^T \omega \vec{x}_2 = 0 \Rightarrow \left(\vec{h}_1 - i\vec{h}_2\right)^T \omega \left(\vec{h}_1 - i\vec{h}_2\right) = 0$

The image of the Absolute Conic Ω_∞: ω

- Therefore, we have:
  - $\vec{h}_1^T \omega \vec{h}_1 - \vec{h}_2^T \omega \vec{h}_2 + i\left(\vec{h}_1^T \omega \vec{h}_2 + \vec{h}_2^T \omega \vec{h}_1\right) = 0$, and
  - $\vec{h}_1^T \omega \vec{h}_1 - \vec{h}_2^T \omega \vec{h}_2 - i\left(\vec{h}_1^T \omega \vec{h}_2 + \vec{h}_2^T \omega \vec{h}_1\right) = 0$
- Since ω is symmetry we must have:
  - $\vec{h}_1^T \omega \vec{h}_2 = \vec{h}_2^T \omega \vec{h}_1 = 0$ ...(1)
- As a result we also should have
  - $\vec{h}_1^T \omega \vec{h}_1 = \vec{h}_2^T \omega \vec{h}_2$ ...(2)
- Recall that $\vec{h}_1$ and $\vec{h}_2$ are part of the homography obtained by showing one pattern to the camera.
- Thus, we have two equations but 5 unknowns of ω .
- As such, we need at least three probing planes (or three different computed homographies) to estimate ω.

The image of the Absolute Conic Ω_∞: ω

- Methodology:
  - Setup your camera
  - Move at least three planes with respect to the camera
    - `https://docs.opencv.org/3.4/da/d0d/tutorial_camera_calibration_pattern.html`
  - That's all !!!!
- **Question: Could we just show one probe plane to the camera and calibrate it?**
  - Ans: Yes, provided your probe plane is rich with patterns.
    - It is a Research Area.

Camera Calibration: Zhang's Algorithm

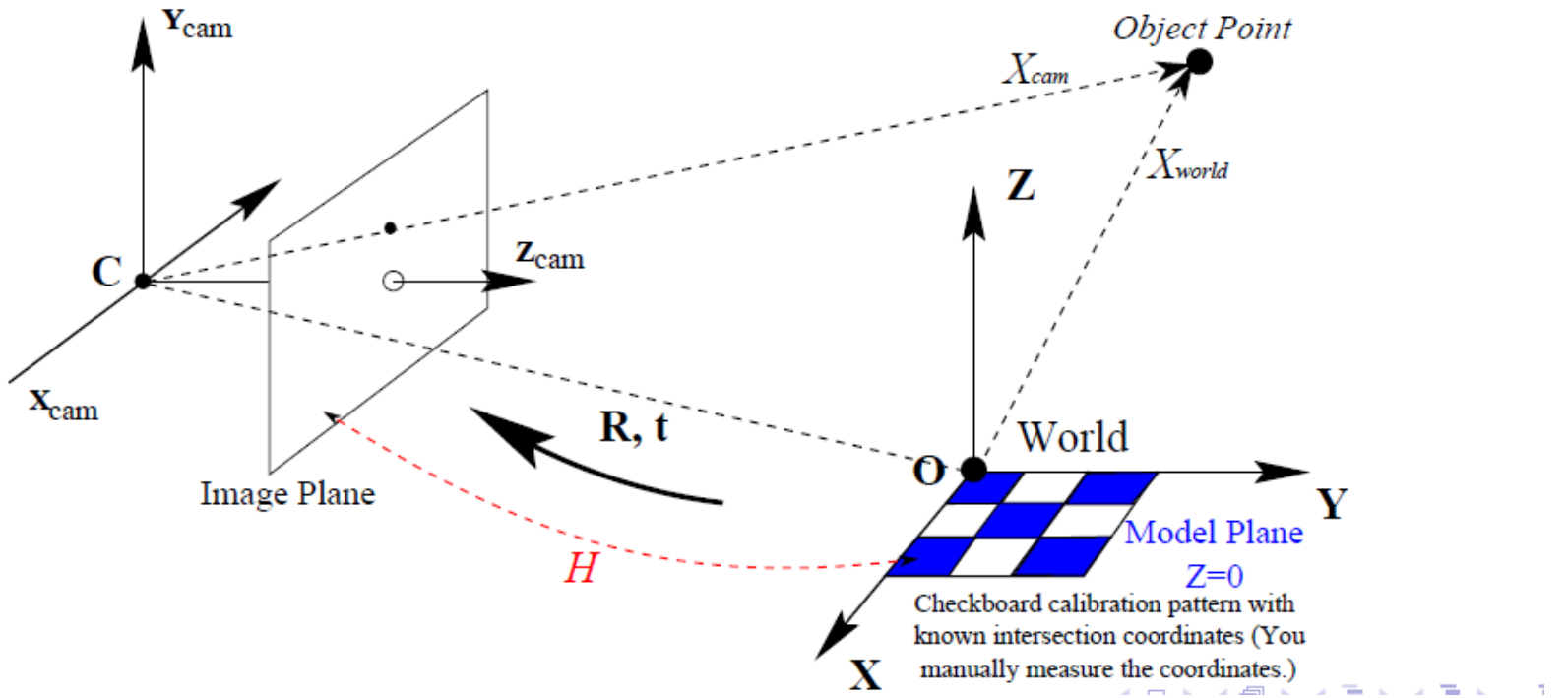
- Recall camera imaging:
  - $\vec{x} = KR\left[I|-\vec{C}\right]\vec{X}_{world}$ or $\vec{x} = K\left[R|\vec{t}\right]\vec{X}_{world}$
  - $\vec{x} = K\left[I|\vec{0}\right]\vec{X}_{cam}$
- Note that $\vec{X}_{cam} = R\vec{X}_{world} + \vec{t}$, where $\vec{t} = -R\vec{C}$
- Therefore, In a matrix form we can write $\vec{x} = K\left[I|\vec{0}\right]\begin{bmatrix} R & \vec{t} \\ \vec{0}^T & 1 \end{bmatrix}\vec{X}_{world}$

Camera Calibration: Zhang's Algorithm

- The **camera calibration algorithm** developed originally by **Zhengyyou Zhang** (currently leading both the Robotic X Lab and AI Lab at Tencent after 20 years of work at Microsoft Research) on the basis of the ideas discussed in the previous lecture (or in the previous slides) **is the world's most popular such algorithm today.**

Zhengyou Zhang's Calibration Method

- Refer to "**A Flexible New Technique for Camera Calibration**" at `http://research.microsoft.com/en-us/um/people/zhang/calib/`
  - A technique to figure out both of the intrinsic (K) and extrinsic parameters (R, t) of the camera



Main Observation

- $\vec{x} = K \begin{bmatrix} \vec{r}_1 & \vec{r}_2 & \vec{r}_3 & \vec{t} \end{bmatrix} \vec{X}_{world}$
- **Recall that our model (check-board calibration) pattern is in $X - Y$ plane i.e. $Z = 0$ of the world coordinate system**
- Therefore, all of the world points $\vec{X}_{world}$ located on the model pattern will have the following coordinates $\begin{pmatrix} X \\ Y \\ 0 \\ W \end{pmatrix}$
- As such, we will have

$$\vec{x} = K \begin{bmatrix} \vec{r}_1 & \vec{r}_2 & \vec{r}_3 & \vec{t} \end{bmatrix} \begin{pmatrix} X \\ Y \\ 0 \\ W \end{pmatrix} = K \begin{bmatrix} \vec{r}_1 & \vec{r}_2 & \vec{t} \end{bmatrix} \begin{pmatrix} X \\ Y \\ W \end{pmatrix} = H \begin{pmatrix} X \\ Y \\ W \end{pmatrix}, \text{ where}$$

$$\boldsymbol{H = K \begin{bmatrix} \vec{r}_1 & \vec{r}_2 & \vec{t} \end{bmatrix}}$$

Key Idea

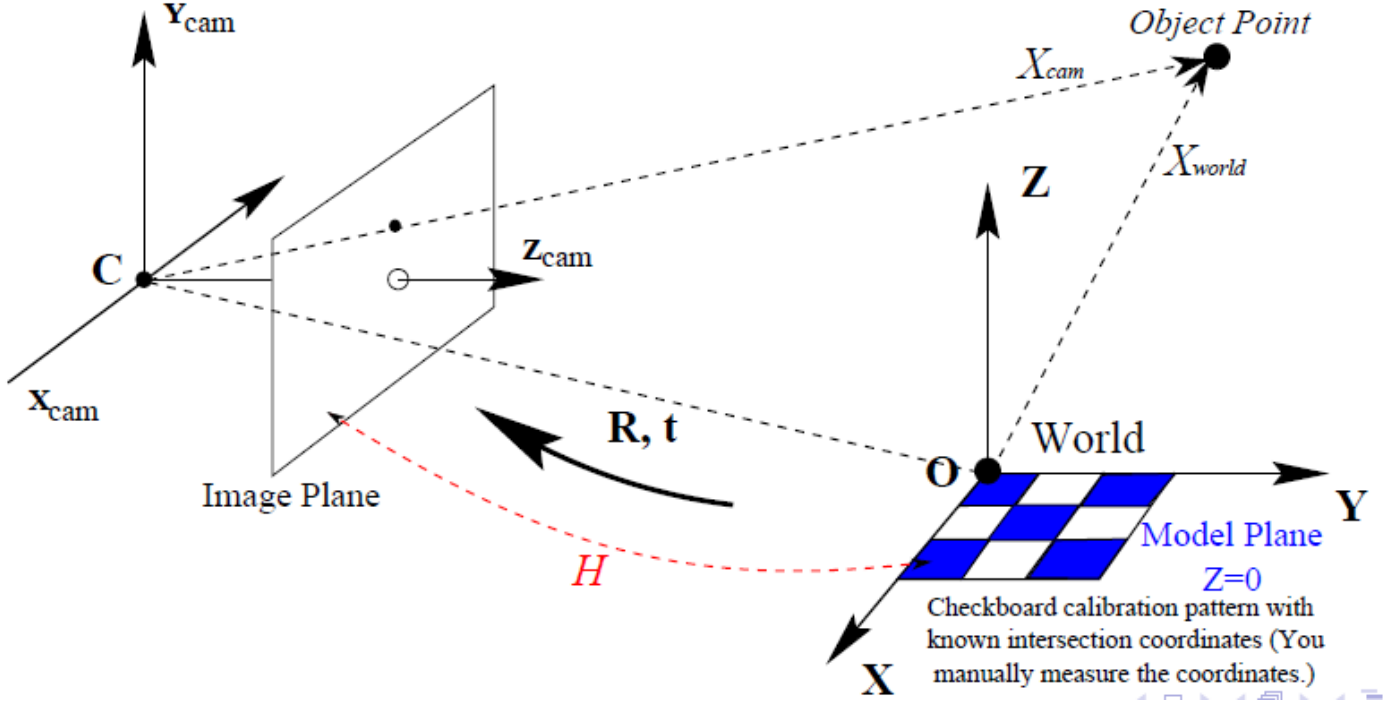
- So, for each view of the camera for the calibration pattern, we can estimate H using four non-colinear point correspondences.

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} = [\vec{h}_1 \quad \vec{h}_2 \quad \vec{h}_3]$$

$$H = K\,[\vec{r}_1 \quad \vec{r}_2 \quad \vec{t}]$$

$$\vec{r}_1 = K^{-1}\,\vec{h}_1$$
$$\vec{r}_2 = K^{-1}\,\vec{h}_2$$
$$\vec{t} = K^{-1}\,\vec{h}_3$$

$$\vec{r}_3 = \vec{r}_1 \times \vec{r}_2$$



Matlab: Camera Calibration

- Please review the Matlab Camera Calibration Documentation at:
  - https://www.mathworks.com/help/vision/camera-calibration.html

Demo Using Matlab